



COMSATS University Islamabad

Department of Computer Science

Course Information

Course Code: **CSC354**

Credit Hours: **3(3,0)**

Lab Hours/Week: **0**

Course Title: **Machine Learning**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Text and Reference Books

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

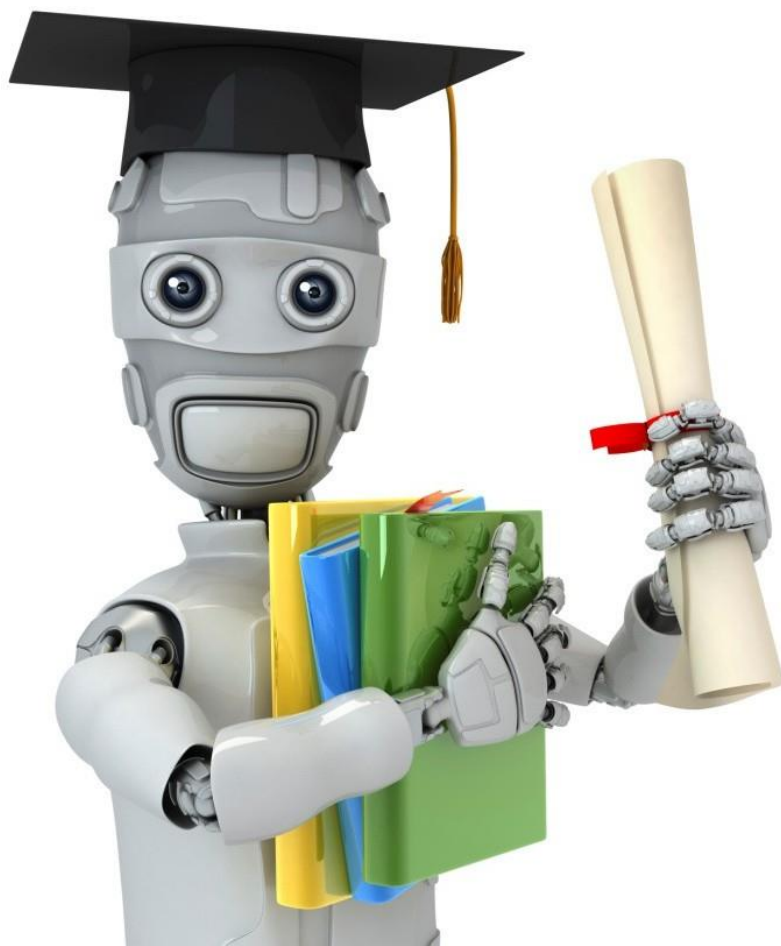
Reference Books:

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, ¹Aurelien Geron, O'Reilly Media, 2017.
Zeena Zulfiquar
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.



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- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on **(zeenatzulfiqar@cuiatd.edu.pk)**



Machine Learning

Comparing Classification Algorithms:
Confusion Matrix, and Analysis of
Variance (ANOVA).

Evaluation Matrix

Model Evaluation

Model evaluation is the process of measuring how well a machine learning model performs on unseen data using different metrics.

Confusion Matrix (Classification)

Confusion Matrix (Classification) A **confusion matrix** is a table used to evaluate the performance of a classification model by comparing **predicted labels with actual labels**. It shows the number of:

True Positives (TP)

True Negatives (TN)

False Positives (FP)

False Negatives (FN)

It helps measure how correctly a model classifies data.

ANOVA (Analysis of Variance)

ANOVA (Analysis of Variance) is a statistical method used to compare the **mean performance of multiple models or groups** to check whether the differences between them are **statistically significant or due to random chance**.

It is commonly used to compare different machine learning models.

Example1

Consider the output of a classifier and the corresponding actual labels:

Output (Predicted)

0 1 0 0 1 1 0 1 0 0 1 0 1 1 0 0 1 0 0 1 1 0 0 1 0 1

Actual Labels

1 1 0 0 1 0 0 1 1 0 1 1 1 0 0 0 1 0 1 1 0 0 0 1 1 0

Evaluation Matrix

Step 1: Count TP, TN, FP, FN

We compare each position:

Results:

- **TP (1 predicted, 1 actual) = 6**
- **TN (0 predicted, 0 actual) = 8**
- **FP (1 predicted, 0 actual) = 5**
- **FN (0 predicted, 1 actual) = 7**

Classification Confusion Matrix

	Actual 1	Actual 0
Predicted 1	TP = 6	FP = 5
Predicted 0	FN = 7	TN = 8

Evaluation Matrix for both Classification

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{6 + 8}{26} = \frac{14}{26} = 0.538$$

$$\checkmark \text{Accuracy} = 53.8\%$$

Evaluation Matrix for both Classification

Precision

$$Precision = \frac{TP}{TP + FP}$$

$$Precision = \frac{6}{6 + 5} = \frac{6}{11} = 0.545$$

✓Precision = 54.5%

Evaluation Matrix for both Classification

Recall

$$Recall = \frac{TP}{TP + FN}$$
$$Recall = \frac{6}{6 + 7} = \frac{6}{13} = 0.462$$

✓Recall = 46.2%

Evaluation Matrix for both Classification

F1 Score

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$
$$F1 = \frac{2 \times 0.545 \times 0.462}{0.545 + 0.462}$$
$$F1 \approx 0.500$$

√F1 Score = 50.0%

Evaluation Matrix for both Classification

5. Specificity

$$\text{Specificity} = \frac{TN}{TN + FP}$$

✓ How well negatives are identified

6. Error Rate

$$\text{Error Rate} = \frac{FP + FN}{TP + TN + FP + FN}$$

✓ Misclassification rate

Evaluation Metrics for Regression Problems

Regression Metrics

- MAE
- MSE
- RMSE
- R^2 Score

Regression Evaluation

X	Actual (Y)	Predicted ($\hat{Y}$)
1	10	12
2	20	18
3	30	33
4	40	37

Step 1: Compute Errors

We calculate:

- $\text{Error} = Y - \hat{Y}$
- $|\text{Error}|$ = absolute error
- Error^2 = squared error

X	Y	$\hat{Y}$	Error (Y- $\hat{Y}$)	$ \text{Error} $	Error^2
1	10	12	-2	2	4
2	20	18	2	2	4
3	30	33	-3	3	9
4	40	37	3	3	9

Step 2: MAE (Mean Absolute Error)

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

$$MAE = \frac{2 + 2 + 3 + 3}{4}$$

$$MAE = \frac{10}{4} = 2.5$$

Step 3: MSE (Mean Squared Error)

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$MSE = \frac{4 + 4 + 9 + 9}{4}$$

$$MSE = \frac{26}{4} = 6.5$$

Step 4: RMSE (Root Mean Squared Error)

$$RMSE = \sqrt{MSE}$$

$$RMSE = \sqrt{6.5} \approx 2.55$$